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Robust water meter reading recognition method for complex scenes

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Abstract

We propose a robust water meter reading recognition method for complex scenarios. In our method, the traditional water meter recognition method has been improved so that it can handle more variable real-world scenarios. A WDPDet network is applied in the digit and pointer area detection. The geometric methods is used for rotation correction for subsequent processing. In particular, a data augmentation method is proposed for half-word character recognition that is difficult to handle in the past, which greatly improve network performance. A coarse-grained projection method for pointer recognition is proposed to deal with more complex interference. We have performed a lot of experiments on different devices and scenarios, and have achieved good accuracy.

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1. Introduction

With the deepening of deep learning technology in the field of image recognition and the gradual popularization of 5G technology in society, automatic recognition of water meter readings has become possible. Although smart water meters have been equipped in many modern areas, non-intelligent digital and pointer water meters still exist. For these old-fashioned water meters, manual meter reading is currently used to record readings. On the one hand, manual meter reading is prone to encounter missed reading and wrong reading. These are not conducive to the correct grasp of water consumption by water conservancy companies. On the other hand, many human and financial

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resources are consumed, which is a huge expenditure. Automatic meter reading technology can overcome these difficulties, not only improve the accuracy, but also reduce the cost of the meter reading process. Therefore, the research on the automatic identification technology of water meter reading is very meaningful and urgently needed.

In recent years, many automated meter reading technologies have been proposed. These technologies are mainly divided into two types: (1) New water meters using digital acquisition and transmission functions, such as pulse remote meters, direct reading remote meters, and other new water meters¹. (2) Automatic meter reading technology based on computer vision²⁻⁵. The new water meter is more intelligent, and the water meter reading can be directly transmitted to the database through the embedded device. Correspondingly, the extra costs (such as production and maintenance costs) generated and hinder their promotion. The second method, based on computer vision's automatic meter reading technology, can obtain the original image information of the water meter through a mobile phone without changing the existing water meter structure, and then use vision technology to automatically identify its reading. The latter has a simple process and low cost, which is of more significance in comparison.

Although the existing computer vision automatic meter reading technology has a relatively high accuracy rate, most of its methods can only show good performance in an experimental environment, and it is still very difficult to apply it to complex and changeable real scenes. In view of this phenomenon, we propose a method based on deep learning that can uniformly identify multiple water meters in complex scenes and consistently has a high accuracy rate. The architecture is shown in Fig. 1. This method first uses the object detection technology to detect the water meter reading area, and then performs rotation correction to determine the type of water meter. Different recognition algorithms are used according to different types of water meters, and the results are finally obtained and saved.

This method uses different device to experiment with different water meter types in different environments, and has obtained high accuracy. The results show that the method has high robustness and can be applied in real scenes.

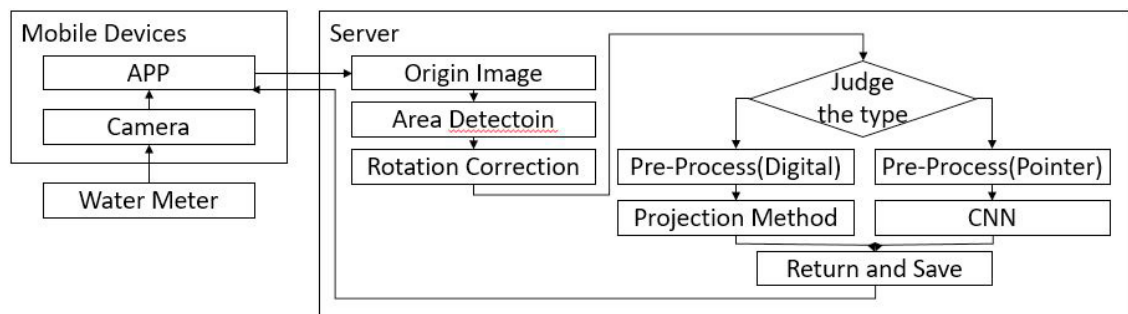


Fig. 1. Water meter reading recognition system architecture

2. Related Work

2.1. Object detection

Object detection is a basic task in computer vision⁶⁻⁸, which locates a object from an image. The current water meter image recognition mostly uses traditional methods⁹⁻¹¹ to locate numbers and pointer areas, which can only locate the target area under limited circumstances, and cannot face more complicated environments. So we applied the deep learning technology to the water meter recognition task and detected the numbers and pointer areas.

2.2. Half-word recognition

An improved halfword recognition algorithm¹² was proposed to significantly reduce the confusion of halfword characters. This method starts from the template matching algorithm and uses the height of the halfword characters as a matching factor. When both the upper and lower halfword characters appear clearly, only the longer end is retained, and then the similarity between the part and the template is considered. BP network¹³ was proposed to recognize, but it was limited to the development of neural network, the method can only deal whole words.

2.3. pointer recognition

The current pointer recognition methods include step size method¹⁴, circumferential gray detection method, and multi-feature template matching method. These methods can only be used when the water meter is in good shooting conditions. When the pointer area is disturbed by bubbles, stains, and light spots, it is difficult to correctly recognize the pointer reading. So here is an improved ray projection method.

3. Proposed method

3.1. Digital and pointer area detection

Due to the poor adaptability of traditional methods, the YOLOv3 model based on deep learning is used to detect the numbers and pointer regions. Because the task has a small number of object the scale of the model has been reduced to some extent. In addition, considering its requirement for accuracy, we added the Attention mechanism to its basic structure, which can improve the accuracy with increasing little calculations. Based on the above improvements, we propose a WDPDet (Water meter Digit and Point Detection). The architecture is shown in Fig. 2.

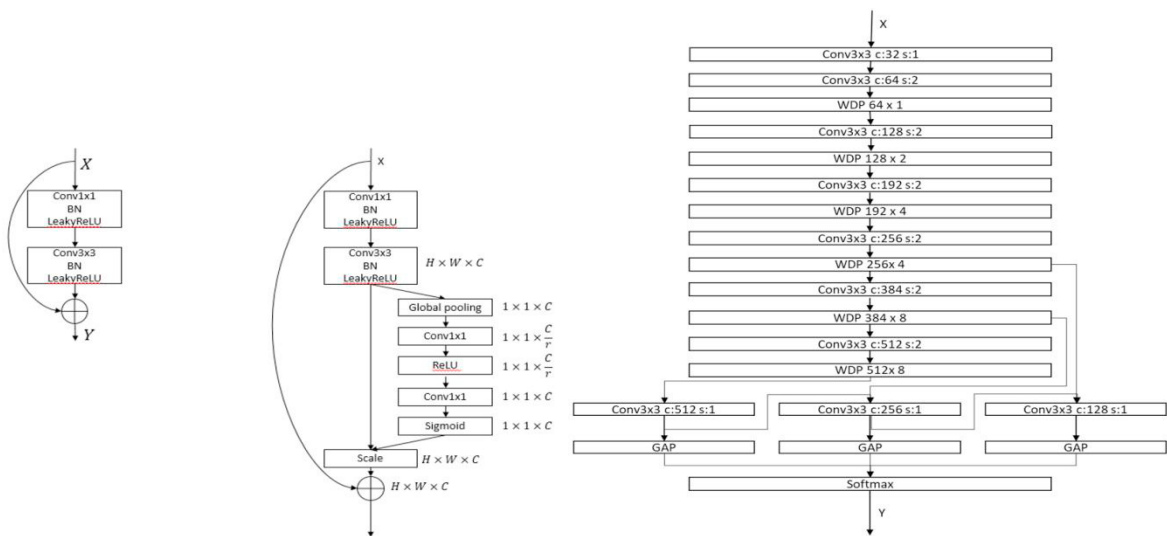


Fig. 2. WDP module and WDPDet

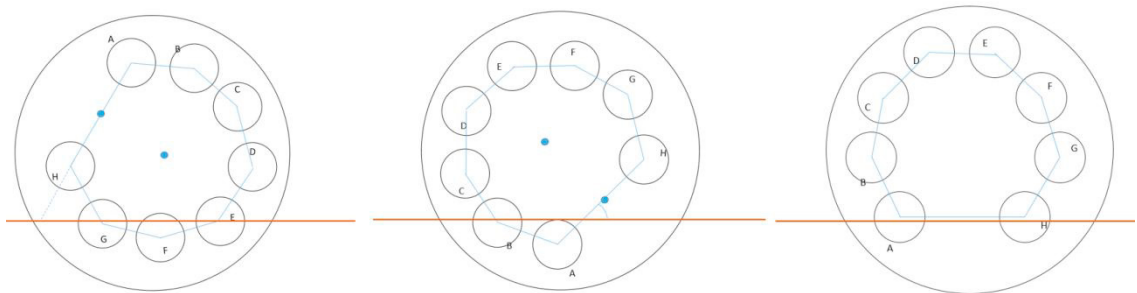


Fig. 3. Rotation correction process

3.2. Rotation correction

Due to the limitation of the actual installation scene of the water meter, many times it is impossible to take a picture of the water meter forward. If you directly identify the unprocessed water meter photos, it will cause errors in the results. At this time, the photo needs to be rotated to the vertical direction, and then the next algorithm is used to identify the angle. Here, we correct the rotation angle based on the coordinates of the pointer area.

We first calculate a convex set based on the center point of the pointer area. It can be found that the longest side AH of the set, which is connected to the highest and lowest bits of the water meter. The edge should be horizontal when the image is in the vertical direction. The edge has an angle with the horizontal line when the image is not vertical. So we rotate the image by the same angle. The specific process is shown in Fig. 3.

3.3. Data augmentation for half-word characters

A segmented single character can be obtained after the process. We use CNN for recognition and find that the recognition rate of full-word characters is much higher than half-word characters. We speculate that because there are fewer halfword data sets, the model cannot effectively learn the characteristics of halfword characters. To solve this problem, we propose a data augmentation method. The specific process is shown in the Fig. 4(a).

First taking n groups of 0-9 numbers, randomly combining them in the vertical direction to form a vertical long image of 0-9, and then moving down a sliding window on the long image; rotate the picture in the sliding window at 0-15 degrees random angles each time to generate the final data image. This method is random in both the upper and lower characters and the rotation angle, which can effectively improve the diversity of the data set. This method not only greatly enriches the data set, but also saves the labor consumption of manually making the data set. Finally, the segmented recognition results are stitched together to obtain a complete reading of the digital area.

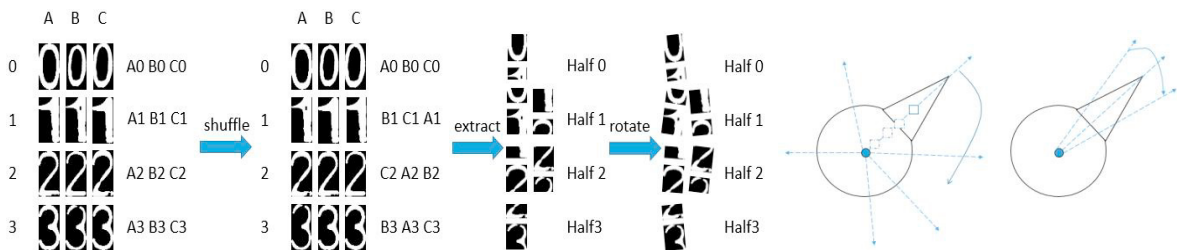


Fig. 4. (a) Data augmentation process (b) Coarse/fine-grained projection process

3.4. Coarse / Fine-grained Projection

For pointer-type water meters, the ray projection method calculates the number of points on the ray from the center of the pointer area to the circumference. The ray with the largest number is the angle line of the pointer. The reading of the pointer can be obtained based on the angle line. However, this method will encounter some problems when used in real scenes. For example, when the pre-processing fails to extract the pointer of the water meter, the calculation of the angle of the rays will be inaccurate. Inspired by the convolution kernel in a convolutional neural network, we propose a coarse-grained projection method that can avoid the above difficulties and increase the speed to a certain extent.

First perform coarse recognition to get a search range, and then perform fine recognition within a given range to get specific readings. Specifically, first use a square matrix similar to the convolution kernel to scan the points on the current ray angle in steps of s . When the proportion of white pixels in the matrix is higher than the threshold, we accumulate the number of white pixels at the current position. Let the longest ray be plus or minus 5 degrees as the search range. Then use the original ray projection method to calculate the final result in the search range in steps of 1. The coarse/fine-grained projection process is shown in Fig. 4(b).

4. Experiment

4.1. Experimental setup

In the experiment, the dataset comes from 40k water meter pictures taken by real people, the size is about 9GB. Some pictures with extremely poor quality are removed, and about 30k pictures are available.

We used YOLO Mark to get the label for object detection on 10k images. In the training of the object detection network, we used the WDPDet model, using 9k images in the dataset for training, and 1k images for testing.

For digital recognition we use VGG as the backbone. We prepared 200 basic pictures for 10 whole-word characters, and then used the proposed data augmentation method to generate several pictures for each whole-word and half-word characters, 90% of which were used as training sets and 10% were used Test set.

Neural network models are written using TensorFlow and Keras, the server GPU is 1070Ti, the Adam optimizer is used, and the initial learning rate is set to the default value.

4.2. Precision Experiment

The test set consist of 1000 pictures, 682 are water meters with 1 digital area and 4 pointer areas, and 318 are water meters with 8 pointer areas, these data

There are 682 digital areas and 5272 pointer areas. We tested 1,000 pictures and only judged whether the correct area was detected. The accuracy obtained is shown in the Table 1.

Table 1. Area detection accuracy

Method	Digital Area acc	Pointer Area acc	average acc
hough	72.58%	68.46%	68.93%
Yolo	95.60%	93.11%	93.40%
WDPDet	97.80%	96.22%	96.38%

We studied the impact of data augmentation on digital recognition. First, we do not use the data augmentation as a baseline. Then using the proposed data augmentation method, four sets of augmentation experiments were performed, and each number was augmented to 1000, 2000, 5000, and 10000. The results are shown in Table 2.

Table 2. Digital recognition accuracy

Method	whole word acc	half word acc	average acc
baseline	95.58%	84.70%	90.14%
aug to 1000	98.31%	92.43%	95.37%
aug to 2000	99.17%	97.59%	98.38%
aug to 5000	99.94%	99.41%	99.68%
aug to 10000	99.94%	99.50%	99.72%

It was found in experiments that the accuracy of recognition of half-word characters is low, only 84.70% when no augmentation is used; after augmentation, the accuracy is significantly increased. When each number is increased to 5000, the effect is basically satisfactory. When expanded to 10000 sheets, the accuracy of whole-word characters no longer changes, while half-word characters only increase slightly. So we ended up using 5000 as the number of augmentation.

This experiment fully illustrates the effectiveness of the data augmentation method, which can greatly improve the accuracy of half-word character recognition.

For the proposed pointer recognition method, we changed the rotation step r and sliding step s of the coarse-grained ray projection stage and performed a series of comparative experiments. The experimental results obtained are shown in the Table 3. It can be seen that when the rotation step is larger and the sliding step is larger, the

accuracy is lower. As the sliding and rotating steps decrease, the accuracy will gradually increase. However, when the experimental step is the smallest ($r = 1$ $s = 1$), it is not as good as $r = 5$ $s = 5$. We speculate that after the step is set too small, it will be more susceptible to interferences such as stains and shadows on the dial. The effect of the error makes the kernel calculation error. So we finally set the rotation step to 5 and the sliding step to 5.

Table 3. Pointer recognition accuracy

Method	s=10	s=5	s=1
r=10	60.72%	69.41%	72.64%
r=5	82.94%	93.57%	92.62%
r=1	83.77%	91.43%	92.95%

We connected the entire system together and performed an overall identification of the data set, and obtained the above results. For multiple models of water meters can be effectively identified under different tilt and rotation angles. The overall results are in Table 4 and some examples are in Figure 5.

Table 4. Overall results accuracy

Method	acc
WDPDet	76.87%
WDPDet + aug to 5000	82.59%
WDPDet + aug to 5000 + CFP	94.12



Fig. 5. Reading recognition results

5. Conclusion

In this paper, we propose a method based on deep learning that can uniformly identify multiple water meters in complex scenes, and consistently has a high accuracy rate. Among them, the WDPDet network model shows significantly better detection performance than traditional methods. The rotation correction method enables the model to process rotated pictures. The augmented data of half-word characters effectively improves the recognition rate of half-word characters. (CFP) also has obvious anti-interference effect. Based on the above methods, a complete set of water meter identification methods has been formed. This method uses different equipment to experiment with different types of water meters under different environments, and has obtained high accuracy results. The results show that the method has high robustness and can be applied in real scenarios.

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References

1. Jianrong, H., On the Selection of Intelligent Remote Transmission Table. *City and Town Water Supply*, 2014(03): p. 64-65.
2. Li Yundong, S.C., Research on Real Time Water Meter Recognition System Based on BP Network. *Computer Measurement & Control*, 2013.
3. Neng, M.Y.X.K., Application of depth convolutional neural network in watermeter digit recognition. *Water Technology*, 2017.11(05): p. 54-57.
4. Yunze Gao, Automatic Watermeter Digit Recognition on Mobile Devices. *Internet Multimedia Computing and Service*, 2018. vol 819.
5. Yu lin, IMAGE TYPE WATER METER CHARACTER RECOGNITION BASED ON EMBEDDED DSP. *ArXiv*, 2015. 1508.06725.
6. Wei Liu, D.A.D.E., SSD: Single Shot MultiBox Detector. *Computer Vision and Pattern Recognition*, 2016.
7. Qin, Z., et al., ThunderNet: Towards Real-time Generic Object Detection. *Computer Vision and Pattern Recognition*, 2019.
8. Redmon, J. and A. Farhadi, YOLOv3: An Incremental Improvement. *Computer Vision and Pattern Recognition*, 2018.
9. Hu, B., The Research and Application of Pointer Water-Meter Recognition System Based on Hough Transform. 2016.
10. YUAN Weiqi, L.H., Study of Remote Reading System of Pointer Water-meter. *Modern Electronics Technique*, 2008(10): p. 61-63+66.
11. Na, C., The study used automatic meter reading on system which based in machine vision. 2011.
12. Kunlun, Application of improved half-word recognition method in automatic meter reading system. *Water Technology*, 2015. 9(01): p. 43-45.
13. LI Yao, Character recognition algorithm adapt to specific kind of water meter. *Computer Engineering and Design*, 2009
14. Sun, J., Research on Water Meter Reading Recognition. 2016, Beijing Institute of Technology.